GENERALIZED HERMITE–BIRKHOFF INTERPOLATION OF ARBITRARY ORDER IN ANY DIMENSION VIA SINGULAR-WEIGHT MOVING LEAST SQUARES

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Abstract. Singular-weight interpolating moving least squares (SWIMLS) reproduces function values at scattered nodes. We study the method when each node carries an arbitrary finite set of linear differential data $q(D)f(x_i)$. We prove that the interpolant is C^∞, give an exact representation of the error in every datum, and deduce that a datum is reproduced exactly if (i) every derivative of its symbol q lies in the span of the symbols prescribed at that node and (ii) $\Delta^{\alpha/2}q = 0$, where α is the (even) order of the weight singularity. Condition (ii) cannot be dropped. Whether condition (i) is needed depends on how the fit to the other nodes moves near x_i: we identify configurations in which it is not needed, and give the error explicitly when (i) fails only at first order. For classical Hermite data of order k, all data are reproduced if and only if $\alpha > k$ (the necessity under a mild condition on the other nodes), which proves a sharpened form of a conjecture of Komargodski and Levin. We also give exact counterexamples, in one and two dimensions, to their claim that derivatives prescribed at points without function values are interpolated, and a completion procedure that makes such data exact. A stable evaluator and experiments in one to three dimensions confirm the theory, and in a two-dimensional study the root-mean-square error decays at rates close to h^{m+1} for local degree m.

1. Introduction

Moving least squares (MLS) builds an approximant point by point: at each evaluation point x one fits a polynomial to the data by weighted least squares, with weights that decay away from x, and evaluates the fit at x. The method goes back to McLain [13, 14] and Lancaster and Šalkauskas [9]; its approximation theory is due to Levin [10] and Wendland [23], see also [5]. When the weights are singular at the nodes, $w_i(x) = |x - x_i|^{-\alpha}$, the approximant interpolates the data [9]; with constant local polynomials it is Shepard's method [22]. Interpolating MLS needs no mesh and no global linear system, and it is local, smooth and exact on polynomials of the local degree; MLS is also optimal in the sense of Backus and Gilbert [2], and nearly interpolating weights are used to impose boundary conditions in meshless methods [17].

In many applications the data include derivatives: slopes and curvatures in curve and surface design, gradients in derivative-enhanced surrogate models, fluxes and normal derivatives in meshless discretisations of differential equations. Kernel methods handle

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such Hermite–Birkhoff data by building the derivatives into the kernel matrix [24, 18], at the price of a global system. In meshless methods, derivative data have been built into MLS fits [1], and generalized MLS approximates derivatives and other functionals from values [16]. For interpolating MLS, Komargodski and Levin [8] proposed using the same singular weights for value and derivative data at a node. They proved that this interpolates values and first derivatives in one dimension, and reported that for data up to second derivatives $\alpha = 2$ is not enough while $\alpha = 4$ works. They conjectured that $\alpha = 2k$ suffices for derivatives up to order k, and stated that the method also interpolates directional derivatives prescribed at points that carry no function value.

Polynomial Hermite–Birkhoff interpolation in several variables is notoriously delicate: whether a data pattern is poised depends on the pattern and on the geometry of the nodes [12, 7, 11]; the natural class of local data is that of D-invariant spaces [4, 19], which characterise ideal interpolation [3, 21]. Interpolating MLS avoids the global poisedness question, since it only asks that the data be unisolvent on the local polynomial space. It replaces it by a local question: which data are reproduced? This paper answers that question.

We call the method SWIMLS (singular-weight interpolating moving least squares) and analyse it for arbitrary finite sets of linear differential data $q(D)f(x_i)$ at each node, in any dimension. The analysis rests on two elementary tools, the Fischer pairing of a symbol with a polynomial and a structure theorem for the local fit near a node, and it leads to an exact formula for the error in each datum.

Contributions.

(1) A structure theorem for SWIMLS near a node (Theorem 4.1): the interpolant is C^∞, and an exact representation of the error in any differential datum (Theorem 4.3).
(2) A sufficient condition for exact reproduction of a datum $q(D)f(x_i)$: closure of the symbol under differentiation and $\Delta^{\alpha/2}q = 0$ (Corollary 4.4). The second condition cannot be dropped: it is necessary at leading order (Proposition 4.7).
(3) The role of the first condition: it is not needed when the minimiser of the "frozen" problem at the other nodes does not move near x_i (Theorem 4.9), e.g. for a single other node or locally proportional weights; when it fails only at first order, the error is given explicitly by Proposition 4.12.
(4) For classical Hermite data of order k: all data are reproduced when $\alpha > k$, and this is sharp (Corollaries 4.5 and 4.8); when $\alpha \leq k$ the error is the linear functional $q \mapsto \left[\Delta^{\alpha/2}q, \tau\right]_{x_i}$ of the datum. This proves, for $k \geq 3$ in a sharper form, the conjecture of [8, §3] that $\alpha = 2k$ suffices.
(5) Exact counterexamples in one and two dimensions to the extension claimed in [8, §3] (Propositions 5.1 and 5.2), and a completion procedure that makes data with gaps exact (Proposition 5.3).

(6) Numerical experiments in one, two and three dimensions that confirm each of these statements datum by datum, based on a numerically stable evaluator of SWIMLS.

Outline. Section 2 shows the method and both failure mechanisms on a one-dimensional example. Section 3 fixes notation, Section 4 contains the theory, and Section 5 treats derivatives at points without values. Section 7 reports the experiments and Section 8 lists open problems. Appendices A and B describe the numerical evaluator and how exactness was verified.

2. A one-dimensional example

Take the nodes $x_i = i$, $i = 0, 1, 2, 3$, prescribe values f_i and slopes f_i', and fix the exponent $\alpha = 2$. At every point x that is not a node, let p_x be the quadratic polynomial that minimises

$$J_x(p) = \sum_{i=0}^{3} \frac{\big(p(x_i) - f_i\big)^2 + \big(p'(x_i) - f_i'\big)^2}{(x - x_i)^2}, \tag{1}$$

and set $u(x) := p_x(x)$. The minimiser is linear in the data, so $u = \sum_i \big(f_i\varphi_i + f_i'\psi_i\big)$ with shape functions φ_i, ψ_i; two of them are shown in Figure 1(a). They have the cardinal property of Hermite interpolation: $\varphi_i(x_j) = \delta_{ij}$, $\varphi_i'(x_j) = 0$, $\psi_i(x_j) = 0$ and $\psi_i'(x_j) = \delta_{ij}$. Nothing in (1) uses the ordering of the nodes or a mesh, and the same definition works with any number of nodes in any dimension.

Why are the slopes reproduced? Near the node $x_1 = 1$, multiply (1) by $(x-1)^2$. The first term becomes $(p(1) - f_1)^2 + (p'(1) - f_1')^2$ and the others are multiplied by the small factor $t = (x-1)^2$. Write the quadratic as $p = p^* + r + c\,(y-1)^2$, where p^* is the line with value f_1 and slope f_1' at 1 and r is a linear polynomial: the node data see r but not c. Balancing the two parts of the functional gives $r = O(t)$, while c is determined by the other nodes. Section 4 makes this precise:

$$p_x(y) = p^*(y) + \nu(x)\,(y-1)^2 + (x-1)^2\,\sigma(x)(y),$$

with smooth coefficients ν and σ. At $y = x$ both correction terms vanish to second order at $x = 1$, so $u(1) = f_1$ and $u'(1) = f_1'$.

The same picture explains what can go wrong. If we also prescribe second derivatives and use local cubics, the correction $(x-1)^2\sigma(x)(x)$ contributes $2\sigma(1)(1)$ to $u''(1)$, and the datum $f''(1)$ is missed (Figure 1(b): $u''(1) = 1.05$ against $f''(1) = -3.04$). With $\alpha = 4$ the correction is $(x-1)^4\sigma(x)(x)$, which does not affect the second derivative, and $f''(1)$ is reproduced. This is the rule "α larger than the order of the data" of Corollary 4.5.

The other failure concerns gaps in the data. Prescribe values at 0 and 2 and slopes at 1 and 3 (Figure 1(c)). At $x_1 = 1$ the only datum, the slope, does not see constants. The component $\nu_0(x)$ of p_x along the constants is then chosen by the other nodes, it varies with x, and its derivative $\nu_0'(1)$ adds to $u'(1)$. The prescribed slope 0.306 comes

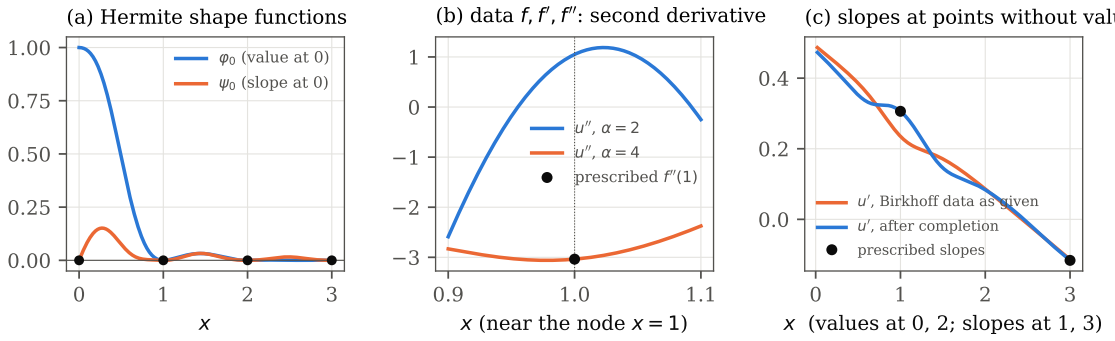


FIGURE 1. The one-dimensional example. (a) Shape functions for the value and the slope at $x = 0$ (nodes $0, 1, 2, 3$, $m = 2$, $\alpha = 2$). (b) Data f, f', f'' at the nodes: u'' near the node $x = 1$ for $\alpha = 2$ and $\alpha = 4$; only $\alpha = 4$ reproduces $f''(1)$ (Corollary 4.5). (c) Values at $0, 2$ and slopes at $1, 3$, $m = 2$: the slopes are not reproduced (compare Proposition 5.1) until the data are completed (Proposition 5.3).

out as 0.235. Adding an estimate of the missing values $f(1)$ and $f(3)$ to the data, by any method, restores exactness (Proposition 5.3); the slopes are then reproduced to the accuracy of finite differences.

3. Setting and notation

3.1. **Polynomials, symbols and the Fischer pairing.** For $d \geq 1$ and $m \geq 0$ let $\Pi_m = \Pi_m(\mathbb{R}^d)$ be the real polynomials of total degree at most m. Multi-indices are $\beta \in \mathbb{N}_0^d$, with $|\beta| = \sum_j \beta_j$, $\beta! = \prod_j \beta_j!$, $D^\beta = \partial^{|\beta|}/\partial x_1^{\beta_1} \cdots \partial x_d^{\beta_d}$. A polynomial $q(\xi) = \sum_\beta q_\beta \xi^\beta$, called a *symbol*, defines the differential operator $q(D) = \sum_\beta q_\beta D^\beta$. We write $\partial^\gamma q$ for derivatives of a symbol with respect to ξ, and $\Delta q = \sum_j \partial_j^2 q$.

For a symbol q, a smooth function g and a point z we set

$$[q, g]_z := \big(q(D)g\big)(z), \qquad [q, g] := [q, g]_0. \tag{2}$$

For polynomials q, p, $[q, p] = \sum_\beta \beta!\, q_\beta p_\beta$ is the *Fischer inner product* [6, 20]; it is positive definite on Π_m. Four elementary facts will be used.

Lemma 3.1 (Fischer adjointness)**.** *For symbols q and polynomials p, $[q, z_j p] = [\partial_j q, p]$. Consequently $[q, |z|^{2\ell} p] = [\Delta^\ell q, p]$ for every $\ell \in \mathbb{N}_0$, where $|z|^2 = \sum_j z_j^2$.*

Proof. With e_j the j-th unit multi-index, $[q, z_j p] = \sum_\kappa \kappa!\, q_\kappa p_{\kappa - e_j} = \sum_\lambda (\lambda + e_j)!\, q_{\lambda + e_j} p_\lambda = \sum_\lambda \lambda!\, (\lambda_j + 1) q_{\lambda + e_j} p_\lambda = [\partial_j q, p]$. Multiplication by $|z|^2 = \sum_j z_j \cdot z_j$ is therefore adjoint to $\sum_j \partial_j^2 = \Delta$, and the second statement follows by induction. □

Lemma 3.2 (Taylor expansion of operators)**.** *Let $A = (A_1, \dots, A_d)$ and $B = (B_1, \dots, B_d)$ be commuting operators. For every symbol q, $q(A+B) = \sum_\gamma \frac{1}{\gamma!} A^\gamma (\partial^\gamma q)(B)$ (a finite sum).*

Proof. This is Taylor's formula $q(a+b) = \sum_\gamma \frac{a^\gamma}{\gamma!}(\partial^\gamma q)(b)$ for the polynomial q, an identity in the commutative polynomial ring in a, b. □

Lemma 3.3. *Let $\ell \in \mathbb{N}_0$ and $\beta \in \mathbb{N}_0^d$. Then $\Delta^\ell \xi^\beta = 0$ if and only if $\sum_j \lfloor \beta_j/2 \rfloor < \ell$.*

Proof. By the multinomial formula, $\Delta^\ell \xi^\beta = \sum_{|\kappa|=\ell} \frac{\ell!}{\kappa!}\, \partial^{2\kappa}\xi^\beta$, and $\partial^{2\kappa}\xi^\beta = \frac{\beta!}{(\beta-2\kappa)!}\xi^{\beta-2\kappa}$ if $2\kappa \le \beta$ and 0 otherwise. Distinct κ give distinct monomials and all coefficients are positive, so $\Delta^\ell \xi^\beta \neq 0$ if and only if some κ with $|\kappa| = \ell$ satisfies $2\kappa \le \beta$, i.e. $\sum_j \lfloor \beta_j/2 \rfloor \ge \ell$. □

Lemma 3.4. *Let $\alpha \in 2\mathbb{N}$. Then $\sum_{|\gamma|=\alpha} \frac{1}{\gamma!}\big(D^\gamma |x|^\alpha\big)(0)\, \partial^\gamma = \Delta^{\alpha/2}$ as operators on symbols.*

Proof. $|x|^\alpha = \sum_{|\kappa|=\alpha/2} \frac{(\alpha/2)!}{\kappa!} x^{2\kappa}$, and $(D^\gamma x^{2\kappa})(0) = (2\kappa)!$ if $\gamma = 2\kappa$ and 0 otherwise. Hence the left side equals $\sum_{|\kappa|=\alpha/2} \frac{(\alpha/2)!}{\kappa!}\partial^{2\kappa} = \big(\sum_j \partial_j^2\big)^{\alpha/2}$. □

3.2. **Data.** Let $\Omega \subset \mathbb{R}^d$ be open and $X = \{x_1, \dots, x_K\} \subset \Omega$ distinct nodes. Node x_i carries symbols $q_{i,1}, \dots, q_{i,n_i}$ of degree at most m and values $y_{i,1}, \dots, y_{i,n_i} \in \mathbb{R}$, i.e. the interpolation conditions

$$\lambda_{i,s}(f) := \big[q_{i,s}, f\big]_{x_i} = \big(q_{i,s}(D) f\big)(x_i) = y_{i,s}, \qquad s = 1, \dots, n_i,\ i = 1, \dots, K. \tag{3}$$

We write $Q_i = \operatorname{span}\{q_{i,1}, \dots, q_{i,n_i}\}$ for the space of symbols at x_i. Every $q = \sum_s c_s q_{i,s} \in Q_i$ has a prescribed value $y_q := \sum_s c_s y_{i,s}$; we say that the datum q is *reproduced* by a function u if $\big[q, u\big]_{x_i} = y_q$. Classical Hermite data of order k correspond to $Q_i = \operatorname{span}\{\xi^\beta : |\beta| \le k\}$; any other choice will be called Hermite–Birkhoff data. A space of symbols is *D-invariant* if it is closed under differentiation.

Assumption 3.5. (A1) $\alpha \in 2\mathbb{N}$ is even, and $w_i(x) = \rho_i(x)\,|x - x_i|^{-\alpha}$ with $\rho_i \in C^\infty(\Omega)$, $\rho_i > 0$.

(A2) For every i the symbols $q_{i,1}, \dots, q_{i,n_i}$ are linearly independent.

(A3) (Unisolvency) If $p \in \Pi_m$ and $\lambda_{i,s}(p) = 0$ for all i, s, then $p = 0$.

Since $\big[\cdot, \cdot\big]_{x_i}$ is non-degenerate on Π_m (it is the Fischer pairing after translation) and $\deg q_{i,s} \le m$, (A2) is equivalent to the linear independence of the functionals $\lambda_{i,1}, \dots, \lambda_{i,n_i}$ on Π_m.

Definition 3.6 (SWIMLS)**.** For $x \in \Omega \setminus X$ let

$$J_x(p) = \sum_{i=1}^{K} w_i(x) \sum_{s=1}^{n_i} \big(\lambda_{i,s}(p) - y_{i,s}\big)^2, \qquad p_x := \operatorname*{arg\,min}_{p \in \Pi_m} J_x(p), \tag{4}$$

and define the interpolant $u(x) := p_x(x)$.

By (A1) and (A3), J_x is a strictly convex quadratic on Π_m for every $x \notin X$, so p_x is unique; it depends smoothly on $x \in \Omega \setminus X$ because the coefficients of J_x do and its Hessian is positive definite. Hence $u \in C^\infty(\Omega \setminus X)$. The behaviour at the nodes is the subject of the next section.

Remark 3.7. Positive smooth factors for the individual functionals of a node can be included: the first block of the Jacobian in the proof of Theorem 4.1 becomes $2L^*W(x)L$, still invertible, and nothing else changes. If some functionals at x_i are *not* given the singular factor, they must be moved from Q_i to the regular part B_x of the proof, and Q_i, N_i refer to the singular ones only.

4. Main results

This section answers one question: when is a datum $q(D)f(x_i)$ reproduced? The answer comes in four steps, which follow the one-dimensional picture of Section 2.

1. Theorem 4.1 describes the local fit p_x near x_i. It splits into a part fixed by the data at x_i, a part $\nu(x)$ that those data cannot see and that the other nodes choose, and a remainder damped by $|x - x_i|^\alpha$.
2. Theorem 4.3 turns this into an exact formula for the error in any datum, with one term for each part.
3. Each term gives a condition for exactness (Section 4.3): the ν-term vanishes if the derivatives of the symbol q are again data at x_i (condition (i)), and the remainder vanishes if $\Delta^{\alpha/2}q = 0$ (condition (ii)). For classical Hermite data this gives the rule $\alpha > k$.
4. Sections 4.4 and 4.5 ask how far each condition is necessary.

Throughout this section we fix a node x_i and write

$$N_i = \{p \in \Pi_m : \lambda_{i,s}(p) = 0,\ s = 1, \dots, n_i\} = \{p \in \Pi_m : \big[q, p\big]_{x_i} = 0\ \forall q \in Q_i\}$$

for the polynomials that the data at x_i do not see.

4.1. Structure near a node.

Theorem 4.1 (Structure near a node)**.** *Let Assumption 3.5 hold, fix an inner product on Π_m and let $R = N_i^\perp$. There are an open neighbourhood $U \subset \Omega$ of x_i with $U \cap X = \{x_i\}$, a polynomial $p_i^* \in R$ with $\lambda_{i,s}(p_i^*) = y_{i,s}$ for all s, and C^∞ maps $\nu : U \to N_i$ and $\sigma : U \to R$ such that*

$$p_x = p_i^* + \nu(x) + |x - x_i|^\alpha\, \sigma(x) \qquad \text{for all } x \in U \setminus \{x_i\}. \tag{5}$$

More precisely, $\nu(x) = n(x, |x - x_i|^\alpha)$ and $\sigma(x) = s(x, |x - x_i|^\alpha)$ for C^∞ maps n, s defined near $(x_i, 0)$, and $Z(x) := n(x, 0)$ is the frozen minimiser

$$Z(x) = \operatorname*{arg\,min}_{n \in N_i} B_x(p_i^* + n), \qquad B_x(p) := \sum_{j \neq i} \frac{w_j(x)}{\rho_i(x)} \sum_{s=1}^{n_j} \big(\lambda_{j,s}(p) - y_{j,s}\big)^2. \tag{6}$$

Applying this at every node, u extends to a function in $C^\infty(\Omega)$, with node value $u(x_i) = \big(p_i^* + Z(x_i)\big)(x_i)$.

Proof. Write $\Lambda_i : \Pi_m \to \mathbb{R}^{n_i}$, $p \mapsto (\lambda_{i,s}(p))_s$, and Λ_j likewise. By (A2) Λ_i is surjective with kernel N_i, so $L := \Lambda_i|_R : R \to \mathbb{R}^{n_i}$ is bijective; put $p_i^* := L^{-1}y_i \in R$. Every $p \in \Pi_m$ is uniquely $p = p_i^* + r + n$ with $r \in R$, $n \in N_i$, and then $\Lambda_i p - y_i = Lr$.

Let U be an open neighbourhood of x_i containing no other node. For $x \in U \setminus \{x_i\}$ divide J_x by $w_i(x) > 0$:

$$\frac{J_x(p)}{w_i(x)} = |Lr|^2 + t(x)\, B_x(p), \qquad t(x) := |x - x_i|^\alpha,$$

with B_x as in (6). Since α is even, $t(x) = \big(\sum_k (x_k - x_{i,k})^2\big)^{\alpha/2}$ is a polynomial in x; and $x \mapsto B_x$ is a C^∞ family of convex quadratics on U, because $\rho_i > 0$ and the w_j, $j \neq i$, are smooth on U. A polynomial $p = p_i^* + r + n$ minimises J_x if and only if the gradient vanishes (the problem is convex), that is

$$2L^*Lr + t(x)\nabla_R B_x(p) = 0, \qquad t(x)\nabla_N B_x(p) = 0,$$

where ∇_R, ∇_N denote the components of the gradient in R and N_i and L^* is the adjoint of L. Since $t(x) > 0$ for $x \neq x_i$ we may substitute $r = t(x)\, s$ and divide by $t(x)$, obtaining $F(x, t(x); s, n) = 0$ for the map $F : U \times \mathbb{R} \times R \times N_i \to R \times N_i$,

$$F(x, t; s, n) := \Big(2L^*Ls + \nabla_R B_x(p_i^* + ts + n),\ \nabla_N B_x(p_i^* + ts + n)\Big),$$

which is C^∞ in all variables, including t near 0.

The point $(x_i, 0)$. Here $F = 0$ reads $\nabla_N B_{x_i}(p_i^* + n) = 0$ and $s = -(2L^*L)^{-1}\nabla_R B_{x_i}(p_i^* + n)$. The restriction of B_x to the affine space $p_i^* + N_i$ is strictly convex for every $x \in U$: its Hessian is $2\sum_{j\neq i} \frac{w_j(x)}{\rho_i(x)} \Lambda_j^* \Lambda_j$ restricted to N_i, with $w_j(x) > 0$ since $x \neq x_j$; and if $n \in N_i$ satisfies $\Lambda_j n = 0$ for all $j \neq i$ then all data functionals vanish on n, so $n = 0$ by (A3). Hence there is exactly one solution (s_0, n_0), and $n_0 = Z(x_i)$; more generally, for every $x \in U$ the equation $F(x, 0; s, n) = 0$ has the unique solution

$$n = Z(x), \qquad s = -(2L^*L)^{-1}\nabla_R B_x\big(p_i^* + Z(x)\big).$$

Implicit function theorem. At $(x_i, 0; s_0, n_0)$ the derivative of F with respect to (s, n) is block triangular,

$$\frac{\partial F}{\partial (s, n)} = \begin{pmatrix} 2L^*L & \big(\nabla^2 B_{x_i}\big)_{RN} \\ 0 & \big(\nabla^2 B_{x_i}\big)_{NN} \end{pmatrix},$$

because every derivative with respect to s of a B-term carries the factor $t = 0$. Both diagonal blocks are invertible (L is injective; the second block is the Hessian above). By the implicit function theorem there are a neighbourhood $U' \times (-\tau_0, \tau_0)$ of $(x_i, 0)$ and a C^∞ map $(x, t) \mapsto (s(x, t), n(x, t))$ with $F(x, t; s(x, t), n(x, t)) = 0$; by the uniqueness just shown, $n(x, 0) = Z(x)$. Shrinking U to $U' \cap \{t(x) < \tau_0\}$, for $x \in U \setminus \{x_i\}$ the polynomial $p_i^* + t(x)s(x, t(x)) + n(x, t(x))$ satisfies the gradient equations of the strictly convex function J_x, hence it equals p_x (only uniqueness of the critical point of J_x is used). Setting $\nu(x) := n(x, t(x)) \in N_i$ and $\sigma(x) := s(x, t(x)) \in R$ gives (5); both maps are C^∞ because t is a polynomial. Finally $u(x) = p_x(x)$ is, by (5), the restriction to $U \setminus \{x_i\}$ of the C^∞ function $x \mapsto (p_i^* + \nu(x) + t(x)\sigma(x))(x)$, whose value at x_i is $(p_i^* + Z(x_i))(x_i)$. □

In words: as $x \to x_i$ the singular weight forces the local fit to match the data at x_i. The part of p_x these data can see converges to p_i^*, with an error of order $|x - x_i|^\alpha$; this is the term $|x - x_i|^\alpha \sigma(x)$. The part they cannot see, $\nu(x) \in N_i$, is decided by the other nodes: at $x = x_i$ it is the polynomial $Z(x_i)$ that fits their data best among all polynomials that match the data at x_i, but in general it moves with x. Both ν and σ are smooth, and this is what makes u smooth at the nodes.

Remark 4.2. The decomposition (5) is canonical up to the choice of the complement R of N_i; the node value $u(x_i)$ and all errors below do not depend on it. Evenness of α is used twice: to make $t(x) = |x - x_i|^\alpha$ smooth at x_i (Theorem 4.1) and to make it a polynomial, so that Lemmas 3.1 and 3.4 apply (Theorems 4.3 and 4.9). If the ρ_i are real analytic, the analytic implicit function theorem shows that u is real analytic. For odd α the function t is not smooth at x_i, our arguments do not apply, and we do not consider this case.

4.2. **An exact error representation.**

Theorem 4.3 (Error representation)**.** *With the notation of Theorem 4.1, let* $\tau(z) := \sigma(z)(z)$ *for* $z \in U$ *(a* C^∞ *scalar function). For every symbol* q*,*

$$[q, u]_{x_i} = [q, p_i^*]_{x_i} + \sum_\gamma \frac{1}{\gamma!} D^\gamma \Big[x \mapsto [\partial^\gamma q, \nu(x)]_{x_i} \Big](x_i) + [\Delta^{\alpha/2} q, \tau]_{x_i}. \tag{7}$$

The $\gamma = 0$ *term of the middle sum is* $[q, \nu(x_i)]_{x_i}$.

Proof. On $U \times \mathbb{R}^d$ consider the C^∞ function $\Phi(x, y) = p_i^*(y) + \nu(x)(y) + t(x)\,\sigma(x)(y) =: \Phi_1 + \Phi_2 + \Phi_3$, which is a polynomial in y and satisfies $u(z) = \Phi(z, z)$ for all $z \in U$, including $z = x_i$ by the definition of the extension. By the chain rule along the diagonal, $D_z^\beta[\Phi(z, z)] = \big[(D_x + D_y)^\beta \Phi\big](z, z)$, hence $[q, u]_{x_i} = \big[q(D_x + D_y)\Phi\big](x_i, x_i)$. We treat the three parts.

Φ_1 does not depend on x, so it contributes $(q(D)p_i^*)(x_i) = [q, p_i^*]_{x_i}$.

For Φ_2, Lemma 3.2 with $A = D_x$, $B = D_y$ gives

$$q(D_x + D_y)\Phi_2 = \sum_\gamma \frac{1}{\gamma!} D_x^\gamma \, (\partial^\gamma q)(D_y)\Phi_2.$$

Setting $y = x_i$ commutes with D_x, and $\big((\partial^\gamma q)(D_y)\Phi_2\big)(x, x_i) = [\partial^\gamma q, \nu(x)]_{x_i}$; setting $x = x_i$ gives the middle term of (7).

For Φ_3 we use the diagonal again: its contribution is $[q, t\,\tau]_{x_i} = \big(q(D)[\,|z - x_i|^\alpha \tau(z)]\big)(x_i)$. Translate x_i to 0 and let $T\tau$ be the Taylor polynomial of τ at 0 of degree $\deg q$. Since $|z|^\alpha(\tau - T\tau)$ vanishes at 0 to order larger than $\deg q$, it is annihilated by $q(D)$ at 0, and by Lemma 3.1

$$[q, |z|^\alpha \tau] = [q, |z|^\alpha T\tau] = [\Delta^{\alpha/2} q, T\tau] = [\Delta^{\alpha/2} q, \tau],$$

the last equality because $\deg \Delta^{\alpha/2} q \le \deg q - \alpha \le \deg T\tau$. □

The three terms of (7) correspond to the three parts of (5). The first is the prescribed value y_q whenever $q \in Q_i$. The second measures the motion of the invisible part ν; it pairs the derivatives $\partial^\gamma q$ of the symbol with $\nu(x)$, and every $\partial^\gamma q$ that is itself a datum at x_i drops out, because the data at x_i do not see N_i. The third comes from the damped remainder, and the singularity of the weight enters only through the operator $\Delta^{\alpha/2}$: for the Fischer pairing, multiplication by $|z|^\alpha$ is adjoint to $\Delta^{\alpha/2}$ (Lemma 3.1).

4.3. **Exact reproduction.**

Corollary 4.4 (Sufficient condition)**.** *Let Assumption 3.5 hold and let $q \in Q_i$. If*

(i) $\partial^\gamma q \in Q_i$ for every multi-index γ, and
(ii) $\Delta^{\alpha/2} q = 0$,

then the datum q is reproduced: $\big[q, u\big]_{x_i} = y_q$. If Q_i is D-invariant, the error is the linear functional

$$\big[q, u\big]_{x_i} - y_q = \big[\Delta^{\alpha/2} q, \tau\big]_{x_i}, \qquad q \in Q_i. \tag{8}$$

Proof. In (7), $\big[q, p_i^*\big]_{x_i} = \sum_s c_s \lambda_{i,s}(p_i^*) = y_q$. By (i), every $\partial^\gamma q$ (including $\gamma = 0$) lies in Q_i, and $\nu(x) \in N_i$, so $\big[\partial^\gamma q, \nu(x)\big]_{x_i} = 0$ for all $x \in U$: the middle term vanishes. This proves (8) when Q_i is D-invariant, and the last term vanishes by (ii). □

Condition (i) concerns the datum and the data at its own node only; it holds for every datum when Q_i is D-invariant, the classical setting of multivariate Hermite interpolation [19, 3]. Condition (ii) holds whenever $\deg q < \alpha$, but also, for instance, for every harmonic symbol when $\alpha = 2$. At a D-invariant node, (8) says more: the error is a *linear functional* of the datum, which only sees $\Delta^{\alpha/2} q$.

Corollary 4.5 (Hermite data)**.** *Let x_i carry classical Hermite data of order $k \le m$, i.e. $Q_i = \operatorname{span}\{\xi^\beta : |\beta| \le k\}$. Then every $q \in Q_i$ with $\Delta^{\alpha/2} q = 0$ is reproduced. In particular:*

(a) the derivative datum $D^\beta f(x_i)$ is reproduced whenever $\sum_j \lfloor \beta_j/2 \rfloor < \alpha/2$;
(b) all data at x_i are reproduced if $\alpha > k$, i.e. for $\alpha = k+1$ when k is odd and $\alpha = k+2$ when k is even;
(c) for $\alpha = 2$ the monomial data covered by (a) are exactly the multilinear ones ($\beta_j \le 1$ for all j, e.g. $\partial_x\partial_y$, $\partial_x\partial_y\partial_z$), whatever their order.

Proof. Q_i is D-invariant, so Corollary 4.4 applies; (a) and (c) are Lemma 3.3 with $\ell = \alpha/2$, and (b) follows from $\deg q \le k < \alpha$. □

Example 4.6. In two dimensions, let x_i carry $f, \partial_x f, \partial_y f, \partial_x^2 f, \partial_x\partial_y f, \partial_y^2 f$ ($k = 2$). For $\alpha = 4$ all six data are reproduced. For $\alpha = 2$, the symbols $1, \xi_1, \xi_2, \xi_1\xi_2$ are harmonic, so the value, the gradient and the mixed derivative are reproduced; $\partial_x^2 f$ and $\partial_y^2 f$ are not, but by (8) both have the same error $\big[2, \tau\big]_{x_i} = 2\tau(x_i)$, so their difference $\partial_x^2 f - \partial_y^2 f$ is reproduced. With third derivatives ($k = 3$, $\alpha = 2$) the same argument gives, e.g., that the error in $\partial_x^3 f$ is three times the error in $\partial_x\partial_y^2 f$, since $\Delta\xi_1^3 = 6\xi_1 = 3\,\Delta(\xi_1\xi_2^2)$.

Corollary 4.5(b) proves the conjecture of [8, §3] that $\alpha = 2k$ suffices for data up to order k (they also observed that $\alpha = 2$ fails and $\alpha = 4$ works for $k = 2$). Their conjecture refers to "an appropriate definition of the inner product", which they do not specify; we read it as the same singular weight for all data at a node, as in their one-dimensional construction. For $k \geq 3$ Corollary 4.5(b) is a strict improvement, e.g. $\alpha = 4$ suffices for $k = 3$.

4.4. **Is condition (ii) necessary?** The next result shows that condition (ii) cannot be dropped, at least for data of degree α, and with it that Corollary 4.5(b) is sharp.

Proposition 4.7 (Necessity of (ii), leading order)**.** *In addition to Assumption 3.5, assume that the data at the nodes other than x_i are unisolvent on Π_m by themselves. Let $q \in Q_i$ satisfy (i), with $\deg q = \alpha$ and $\Delta^{\alpha/2} q \neq 0$. Then, for every choice of the data at x_i, the datum q is reproduced only if the data at the other nodes lie in a fixed affine hyperplane.*

Proof. By (i) the middle term of (7) vanishes, and $\Delta^{\alpha/2} q = c$ is a non-zero constant, so the error is $c\,\tau(x_i) = c\,\sigma(x_i)(x_i) = c\,s_0(x_i)$ with s_0 from the proof of Theorem 4.1. The error is an affine function of the data at the other nodes (with y_i fixed); it vanishes for the data $y_j = \Lambda_j p_i^*$, so it suffices to exhibit data for which it does not.

Since $\deg q = \alpha \geq 2$, some derivative $\partial^\gamma q$ with $|\gamma| = \alpha$ is a non-zero constant, so $1 \in Q_i$ by (i): point evaluation $\ell(p) = p(x_i)$ is a datum at x_i. It vanishes on N_i and is non-zero on Π_m, hence non-zero on R.

Choose $y_j = \Lambda_j(p_i^* + v)$, $j \neq i$, for some $v \in \Pi_m$. Then $B_{x_i}(p) = \langle K(p - p_i^* - v), p - p_i^* - v\rangle$ with $K = \sum_{j \neq i} \frac{w_j(x_i)}{\rho_i(x_i)} \Lambda_j^* \Lambda_j$, which is positive definite on Π_m by the additional assumption. The minimiser n_0 over N_i is characterised by $K(n_0 - v) \perp N_i$, i.e. $K(n_0 - v) \in R$, so $\nabla_R B_{x_i}(p_i^* + n_0) = 2K(n_0 - v)$ and $s_0 = (L^*L)^{-1} K w$ with $w = v - n_0$. As v ranges over Π_m, w ranges over the K-orthogonal complement $W = \{w : Kw \in R\}$ of N_i, so Kw ranges over all of R and s_0 over all of R. Choosing v with $\ell(s_0) \neq 0$ completes the proof. □

Corollary 4.8 (Sharpness)**.** *Under the hypotheses of Proposition 4.7 on the other nodes, let x_i carry classical Hermite data of order $k \leq m$. If $\alpha \leq k$, the datum $\partial_1^\alpha f(x_i)$ is not reproduced for data at the other nodes outside an affine hyperplane. Hence $\alpha > k$ is the exact condition for all Hermite data of order k to be reproduced for all data values.*

Proof. $q = \xi_1^\alpha \in Q_i$ satisfies (i) by D-invariance, $\deg q = \alpha$ and $\Delta^{\alpha/2} \xi_1^\alpha = \alpha! \neq 0$. □

4.5. **When is condition (i) needed?** When (i) fails, the middle term of (7) involves derivatives of ν, that is, it depends on how the fit to the *other* nodes moves as x approaches x_i. If it does not move at all, condition (i) is not needed; if it moves, a failure of (i) at first order produces an explicit error.

Theorem 4.9 (Frozen minimiser)**.** *Let Assumption 3.5 hold and suppose that the frozen minimiser Z of (6) is constant near x_i. Then every $q \in Q_i$ with $\deg q \le \alpha$ satisfies*

$$\big[q, u\big]_{x_i} - y_q = \big[\Delta^{\alpha/2} q, \partial_t n(x_i, 0) + \tau\big]_{x_i}, \tag{9}$$

where $\Delta^{\alpha/2} q$ is a constant. In particular every $q \in Q_i$ with $\deg q < \alpha$, or with $\deg q = \alpha$ and $\Delta^{\alpha/2} q = 0$, is reproduced, whether or not (i) holds.

Proof. By Theorem 4.1, $\nu(x) = n(x, t(x))$ with $n(x, 0) = Z(x)$ constant, so $\nu(x) = Z(x_i) + t(x)\, g(x)$ with $g(x) = \int_0^1 \partial_t n(x, \theta t(x))\, d\theta$ smooth and $g(x_i) = \partial_t n(x_i, 0)$. Since t vanishes at x_i to order α, $D^\gamma[t\, g](x_i) = 0$ for $|\gamma| < \alpha$ and $D^\gamma[t\, g](x_i) = (D^\gamma t)(x_i)\, g(x_i)$ for $|\gamma| = \alpha$. In the middle term of (7): the $\gamma = 0$ term is $\big[q, \nu(x_i)\big]_{x_i} = 0$ because $q \in Q_i$ and $\nu(x_i) \in N_i$; the terms with $0 < |\gamma| < \alpha$ vanish because the constant $Z(x_i)$ is killed by D^γ and $t\, g$ has vanishing derivatives of these orders; the terms with $|\gamma| > \alpha$ vanish because $\partial^\gamma q = 0$; and the terms with $|\gamma| = \alpha$, in which $\partial^\gamma q$ is a constant, add up by Lemma 3.4 to

$$\sum_{|\gamma| = \alpha} \frac{1}{\gamma!} (D^\gamma t)(x_i)\, \big[\partial^\gamma q, g(x_i)\big]_{x_i} = \big[\Delta^{\alpha/2} q, \partial_t n(x_i, 0)\big]_{x_i}.$$

Together with $\big[q, p_i^*\big]_{x_i} = y_q$ and the last term of (7) this gives (9). □

Corollary 4.10. *The frozen minimiser is constant near x_i, and hence Theorem 4.9 applies, in each of the following cases:*

- *(a)* single scale: *$w_j(x)/\rho_i(x) = c(x)\kappa_j$ near x_i for all $j \ne i$, with a positive function c and constants $\kappa_j > 0$; this includes the case of a single other node ($K = 2$) and weights whose germs at x_i are proportional;*
- *(b)* interpolatory restriction: *the functionals $\lambda_{j,s}$, $j \ne i$, that do not vanish identically on N_i restrict to a basis of the dual of N_i (so that the frozen problem can fit them exactly whatever the weights).*

Proof. (a) $B_x = c(x)\hat{B}$ with $\hat{B}$ independent of x, and the minimiser of $c(x)\hat{B}$ over $p_i^* + N_i$ does not depend on $c(x) > 0$. (b) The frozen objective is a positively weighted sum of squared residuals of functionals that are either constant on $p_i^* + N_i$ or, by assumption, can be made zero simultaneously by exactly one $n \in N_i$, which is then the minimiser for every choice of positive weights. □

Example 4.11. These cases are not exotic: for two nodes, gradient data at x_i without a function value are reproduced for $\alpha \ge 2$; in one dimension with $m = 1$, the data $f'(0)$, $f(1)$, $f'(5/2)$ are all reproduced (case (b): only $f(1)$ sees the constants, which form N_i at the slope nodes). A third node in the first case, or a further value datum in the second, destroys the structure, and the data are then no longer reproduced (Section 7).

In general the frozen minimiser moves. When condition (i) fails only at first order, the error is then given explicitly by the following result.

Proposition 4.12 (First-order error)**.** *Let Assumption 3.5 hold and let $q \in Q_i$ with $\deg q < \alpha$, such that $\partial^\gamma q \in Q_i$ for all $|\gamma| \geq 2$. Then*

(10)

$$\big[q,u\big]_{x_i} - y_q = \sum_{j=1}^{d} \big[\partial_j q, \partial_{x_j} Z(x_i)\big]_{x_i}, \qquad \partial_{x_j} Z(x_i) = -H^{-1}\,\partial_{x_j}\big[\nabla_N B_x\big](p_i^* + Z(x_i))\Big|_{x=x_i},$$

where H is the Hessian of B_{x_i} restricted to N_i.

Proof. In (7) the last term vanishes because $\deg q < \alpha$, and the middle terms with $\gamma = 0$ or $|\gamma| \geq 2$ vanish as in Corollary 4.4. For $|\gamma| = 1$, $D_j\big[\partial_j q, \nu(x)\big]_{x_i} = \big[\partial_j q, \partial_{x_j}\nu(x_i)\big]_{x_i}$ and $\partial_{x_j}\nu(x_i) = \partial_{x_j} n(x_i,0) + \partial_t n(x_i,0)\,\partial_j t(x_i) = \partial_{x_j} Z(x_i)$ since $\alpha \geq 2$. Differentiating $\nabla_N B_x(p_i^* + Z(x)) = 0$ gives the formula for $\partial_{x_j} Z$. □

The right-hand side of (10) is an explicit linear functional of the data; for pure power weights ($\rho \equiv 1$) it is a rational function of the node positions. It vanishes in the cases of Corollary 4.10 and for symmetric configurations (e.g. $\partial_{x_1} w_j(x_i) = 0$ for all j), but it is non-zero in the explicit examples of Section 5, and therefore, by analyticity, for node positions in an open dense set for those data patterns.

Conjecture 4.13. *Let $\rho \equiv 1$, let Q_i be D-invariant and the other nodes unisolvent by themselves. If $\Delta^{\alpha/2} q \neq 0$ then, for node positions outside a proper algebraic subset, q is not reproduced for data outside a hyperplane.*

Remark 4.14. Proposition 4.7 proves the conjecture for $\deg q = \alpha$. It is consistent with all experiments of Section 7, including one with pure power weights and only two nodes, where every datum with $\Delta q \neq 0$ fails at $\alpha = 2$ (Section 7.2). Characterising the node configurations for which a failure of (i) causes an error, beyond the first-order formula (10), is open.

5. Derivative data without function values

5.1. **Counterexamples.** Komargodski and Levin [8, §3] extend their Hermite MLS, with weights $|x - z|^{-\alpha}$, $\alpha \in 2\mathbb{N}$, and local spaces containing the constants and the linear functions, to directional derivatives prescribed at points that need not carry function values, and state that the derivative data are interpolated, the proof being "straightforward … by slight modifications" of the co-located case. Condition (i) fails for such data: the symbol of a first derivative differentiates to the constant 1, which is not in Q_i when no value is prescribed at the node. By Theorem 4.9 this does not always cause an error, but it can, and does in the following examples, computed in exact rational arithmetic (scripts in `code/proofs`). In the second example the gradient is given as two directional derivatives at the same point, as [8, §3] allow for gradient data.

Proposition 5.1. *Let $d = 1$, $m = 1$ (local polynomials $a + by$), $\alpha = 2$, $\rho_i \equiv 1$, and prescribe $f(0) = f(2) = 0$ and $f'(1) = f'(3) = 1$. Then*

$$u(x) = \frac{x\,(x-1)(x-2)\,(x^2-4x+5)}{2x^4 - 14x^3 + 37x^2 - 42x + 19},$$

which satisfies $u(0) = u(2) = 0$ but $u'(1) = -1$ and $u'(3) = 31/25$.

Proof. With $w_z(x) = (x-z)^{-2}$ the normal equations of (4) for (a, b) are

$$\begin{pmatrix} w_0 + w_2 & 2w_2 \\ 2w_2 & 4w_2 + w_1 + w_3 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ w_1 + w_3 \end{pmatrix},$$

whose solution gives $u = a + bx$ as stated; the denominator has no real zeros. The remaining values follow by direct computation. □

Proposition 5.2. *Let $d = 2$, $m = 1$, $\alpha = 2$, $\rho_i \equiv 1$, and prescribe $f = 0$ at $(0,0)$, $(2,0)$, $(0,2)$ and $\nabla f(1,1) = (1,1)$. With $s = x_1 + x_2$, $r = x_1^2 + x_2^2$ and $p = x_1 x_2$,*

$$u = \frac{15sr^2 - 68r^2 - 64pr + 128sr + 16ps - 128r - 96p + 48s}{35r^2 - 120sr + 240r + 208p - 240s + 144},$$

which vanishes at the three value points, while $\nabla u(1,1) = (5/9, 5/9)$.

Proof. Exact computation as in Proposition 5.1, with three unknowns (script in `code/proofs`). The denominator tends to $+\infty$ as $|x| \to \infty$, and its minimum, attained on the diagonal near $(0.898, 0.898)$, is $11.21\ldots > 0$. □

The failure persists for $\alpha = 4$ and for $m = 2$ (Section 7.3).

5.2. **Completion.** Given data at x_i with symbol space Q_i, let $\overline{Q}_i$ be the smallest D-invariant space containing Q_i; it has the same maximal degree as Q_i, since differentiation lowers degrees. Choose a basis $\tilde{q}_{i,1}, \ldots, \tilde{q}_{i,r}$ of a complement of Q_i in $\overline{Q}_i$.

Proposition 5.3 (Completion)**.** *Augment the data at x_i by the functionals $\tilde{q}_{i,l}(D)f(x_i)$ with arbitrary values $\tilde{y}_{i,l}$. If Assumption 3.5 holds for the original data, it holds for the augmented data, and every original datum $q \in Q_i$ with $\Delta^{\alpha/2} q = 0$ (in particular every datum of order less than α) is reproduced by the augmented interpolant, whatever the values $\tilde{y}_{i,l}$.*

Proof. The augmented symbols at x_i form a basis of $\overline{Q}_i$, so (A2) holds; (A3) is preserved by adding data. The prescribed value y_q of an original datum is unchanged, because its representation in the augmented basis is unique. Every $q \in Q_i \subset \overline{Q}_i$ satisfies (i) with respect to the D-invariant space $\overline{Q}_i$, and Corollary 4.4 applies. □

The values $\tilde{y}_{i,l}$ influence the accuracy, not the exactness. In the experiments they are estimated by a smooth (non-singular) moving least-squares fit of the given data.

6. Convergence

Sections 4 and 5 concern exactness at the nodes. We now bound the error $u - f$ everywhere, for data sampled from a smooth function f. The natural setting is that of compactly supported weights: they make the problem local, and then a scaling and compactness argument gives the same order of approximation as for classical moving least squares.

6.1. **Compactly supported singular weights.** Let $\psi \in C^\infty([0,\infty))$ satisfy $\psi > 0$ on $[0,1)$ and $\psi = 0$ on $[1,\infty)$, for instance $\psi(s) = \exp\big(1 - 1/(1-s)\big)$ for $s < 1$. For a support radius $\delta > 0$ we use the weights

$$w_j(x) = |x - x_j|^{-\alpha}\, \psi\big(|x - x_j|^2/\delta^2\big), \qquad \alpha \in 2\mathbb{N}. \tag{11}$$

Here $\rho_j(x) = \psi(|x-x_j|^2/\delta^2)$ is C^∞ but vanishes outside the ball $B(x_j,\delta)$, so we replace (A1) and (A3) by

(A1′) $\alpha \in 2\mathbb{N}$, $w_j(x) = \rho_j(x)|x - x_j|^{-\alpha}$ with $\rho_j \in C^\infty(\Omega)$, $\rho_j \geq 0$ and $\rho_j(x_j) > 0$;
(A3′) for every $x \in \Omega$, the data at the nodes x_j with $\rho_j(x) > 0$ are unisolvent on Π_m.

Under these assumptions J_x is still strictly convex for every $x \notin X$, and the proofs of Section 4 apply verbatim once U is chosen so small that $\rho_i > 0$ on U and every ρ_j that is positive at x_i is positive on U: the only place where positivity of the other weights is used, the strict convexity of the frozen problem, then follows from (A3′) at $x = x_i$. In Proposition 4.7, "the other nodes" are those with $\rho_j(x_i) > 0$, and in Corollary 4.10(a) the condition is required only for the j with $w_j \not\equiv 0$ near x_i. With these readings, the general results of Sections 4 and 5 (Theorems 4.1, 4.3 and 4.9, their corollaries, and Propositions 4.7, 4.12 and 5.3) hold for the weights (11); Propositions 5.1 and 5.2 and the statements that use analyticity concern pure power weights.

The data are weighted according to their order. If $q_{j,s}$ is homogeneous of degree $r_{j,s}$, we minimise

$$J_x(p) = \sum_j w_j(x) \sum_{s=1}^{n_j} \delta^{2r_{j,s}} \big(\lambda_{j,s}(p) - y_{j,s}\big)^2. \tag{12}$$

The constant factors $\delta^{2r_{j,s}}$ do not affect any result of Sections 4 and 5 (see the remark after Definition 3.6); they make every term of (12) scale in the same way, which is what the following proof needs.

6.2. **An error bound.** For a ball B and $f \in C^{m+1}(B)$ let

$$|f|_{m+1,B} := \max_{|\gamma|=m+1} \sup_B |D^\gamma f|.$$

We assume:

(H1) *separation:* $|x_i - x_j| \geq \vartheta\delta$ for $i \neq j$;

(H2) *data:* the symbols are homogeneous of degree at most m with coefficient vectors of Euclidean norm one, and the tuple $(q_{j,1},\dots,q_{j,n_j})$ at each node belongs to a fixed compact set $\mathcal{Q}$ of linearly independent tuples (for example, a finite list of data patterns such as Hermite data of order $k \le m$);

(H3) *local norming:* there are $c_0 \in (0,1)$ and C_N such that for every $x \in \Omega$ and every $p \in \Pi_m$

$$\max_{|z|\le 1} |p(x+\delta z)| \le C_N \max\bigl\{\delta^{r_{j,s}} |\lambda_{j,s}(p)| : \ |x_j - x| \le c_0\delta\bigr\}.$$

Condition (H3) is a quantitative form of (A3′): the data near x determine a polynomial, with a constant that does not depend on x or δ.

Theorem 6.1 (Convergence)**.** *Let u be the SWIMLS interpolant with the weights (11) and the functional (12), for data $y_{j,s} = \lambda_{j,s}(f)$ of a function f, and assume (H1)–(H3) (which imply (A1′) and (A3′)). There is a constant C, depending only on d, m, α, ψ, ϑ, c_0, C_N and $\mathcal{Q}$, such that*

$$|D^\beta(u-f)(x)| \le C\,\delta^{m+1-|\beta|}\,|f|_{m+1,B(x,\delta)} \tag{13}$$

for every $x \in \Omega$, the nodes included, every $|\beta| \le m$ and every f whose restriction to $B(x,\delta)$ is in C^{m+1}. In particular, if a family of quasi-uniform node sets is used with $\delta = ch$, h the fill distance, and (H3) holds with c_0 and C_N independent of h, then $\|D^\beta(u-f)\|_{L^\infty(\Omega)} \le C\,h^{m+1-|\beta|}|f|_{m+1,\Omega_\delta}$ with $\Omega_\delta = \Omega + B(0,\delta)$.

The constant comes from a compactness argument and is not explicit. The bound has the same order as for classical moving least squares with value data [10, 23, 25, 15], and it holds near and at the nodes, where the weights are singular. The structure of the proof is the classical one, polynomial reproduction and a bound on a scaled Lebesgue function; what is new is the bound near the nodes, which rests on Theorem 4.1, and the treatment of general differential data.

Proof. Step 1: shape functions and reproduction. The minimiser of (12) is linear in the data, so $u(x) = \sum_{j,s} a_{j,s}(x)\,y_{j,s}$ with shape functions $a_{j,s}$, which are C^∞ on Ω by Theorem 4.1. Each $a_{j,s}$ vanishes, with all its derivatives, where $|x_j - x| \ge \delta$: there w_j vanishes identically in a neighbourhood or, on the sphere $|x_j - x| = \delta$, to infinite order, since ψ is flat at 1. If the data come from a polynomial $p \in \Pi_m$, then $J_x(p) = 0$, so $p_x = p$ and $u = p$. Hence $\sum_{j,s} a_{j,s}(x)\lambda_{j,s}(p) = p(x)$ for all x, and differentiating in x,

$$\sum_{j,s} D^\beta a_{j,s}(x)\,\lambda_{j,s}(p) = D^\beta p(x), \qquad p \in \Pi_m. \tag{14}$$

Step 2: error identity. Fix $x \in \Omega$ and let T be the Taylor polynomial of f of degree m at x. By (14) with $p = T$ and since $D^\beta T(x) = D^\beta f(x)$ for $|\beta| \le m$,

$$D^\beta(u-f)(x) = \sum_{j,s} D^\beta a_{j,s}(x)\,\lambda_{j,s}(f-T).$$

If $|x_j - x| < \delta$ and $q_{j,s} = \sum_{|\gamma|=r} c_\gamma \xi^\gamma$ with $r = r_{j,s}$, then $D^\gamma T$ is the Taylor polynomial of degree $m - r$ of $D^\gamma f$ at x, and Taylor's theorem gives $|D^\gamma(f - T)(x_j)| \leq C_1 \delta^{m+1-r} |f|_{m+1,B(x,\delta)}$ with C_1 depending only on d and m. Since $|c_\gamma| \leq 1$ and the number of multi-indices of order $r \leq m$ depends only on d and m,

(15)
$$|D^\beta(u - f)(x)| \leq C_2\, \delta^{m+1-|\beta|}\, |f|_{m+1,B(x,\delta)}\, \Lambda_\beta(x), \qquad \Lambda_\beta(x) := \sum_{j,s} \delta^{|\beta|-r_{j,s}}\, |D^\beta a_{j,s}(x)|.$$

It remains to bound Λ_β independently of x, δ and the nodes.

Step 3: scaling. Let $z_j = (x_j - x)/\delta$ for the nodes with $|z_j| \leq 1$, and for $p \in \Pi_m$ let $\tilde{p}(z) = p(x + \delta z)$, a bijection of Π_m. Because $q_{j,s}$ is homogeneous, $\lambda_{j,s}(p) = \delta^{-r_{j,s}} \tilde{\lambda}_{j,s}(\tilde{p})$ with $\tilde{\lambda}_{j,s}(\tilde{p}) = (q_{j,s}(D)\tilde{p})(z_j)$, and $w_j(x) = \delta^{-\alpha} \tilde{w}(z_j)$ with $\tilde{w}(z) = |z|^{-\alpha} \psi(|z|^2)$. Hence

$$J_x(p) = \delta^{-\alpha} \sum_j \tilde{w}(z_j) \sum_s \big(\tilde{\lambda}_{j,s}(\tilde{p}) - \delta^{r_{j,s}} y_{j,s}\big)^2.$$

So $\tilde{p}_x$ is the SWIMLS fit, evaluated at the origin, of the *configuration* $\mathcal{C} = (z_j, Q_j)_j$ with data $\delta^{r_{j,s}} y_{j,s}$, where Q_j is the tuple of symbols at x_j, and $u(x) = \tilde{p}_x(0)$. Denote by $\tilde{a}_{j,s}(\mathcal{C})$ the shape functions of this scaled problem at the origin; for configurations with a point at the origin ($x \in X$) they are defined by continuous extension, which exists by Step 5(b). Then $a_{j,s}(x) = \delta^{r_{j,s}} \tilde{a}_{j,s}(\mathcal{C}(x))$, and since moving x translates all z_j at once (near a given x we keep the points of $\mathcal{C}(x)$: points outside the closed unit ball stay outside, and points leaving it have zero weight),

$$\delta^{|\beta|-r_{j,s}} |D^\beta a_{j,s}(x)| = \big|(E^\beta \tilde{a}_{j,s})(\mathcal{C}(x))\big|, \qquad E := \sum_l \nabla_{z_l}.$$

Thus $\Lambda_\beta(x) = \tilde{\Lambda}_\beta(\mathcal{C}(x))$ with $\tilde{\Lambda}_\beta(\mathcal{C}) := \sum_{j,s} |E^\beta \tilde{a}_{j,s}(\mathcal{C})|$, a function of the configuration alone, in which δ no longer appears.

Step 4: the admissible configurations form a compact set. By (H1), a configuration has $n \leq M(d, \vartheta)$ points in the closed unit ball, at mutual distances at least ϑ. By (H2) its symbol tuples lie in $\mathcal{Q}$. By (H3), with $\tilde{p}$ in place of p,

$$\max_{|z|\leq 1} |\tilde{p}(z)| \leq C_N \max\{|\tilde{\lambda}_{j,s}(\tilde{p})| : \ |z_j| \leq c_0\} \qquad (\tilde{p} \in \Pi_m). \tag{16}$$

Let $\mathcal{K}$ be the set of all configurations with these three properties. Each property is preserved under limits. For (16), pass to a subsequence with a fixed number of points, a fixed labelling and a fixed index set $S = \{j : |z_j| \leq c_0\}$; the limit points with $j \in S$ again satisfy $|z_j| \leq c_0$, and $\tilde{\lambda}_{j,s}(\tilde{p})$ depends continuously on z_j and Q_j, so the inequality passes to the limit. Hence $\mathcal{K}$ is compact, as a finite union, over $n \leq M$, of closed and bounded sets.

Step 5: continuity. We show that every $\mathcal{C}_0 \in \mathcal{K}$ has a neighbourhood, in the space of all (z_j, Q_j) with independent tuples Q_j, on which all $\tilde{a}_{j,s}$ are C^∞ functions of $(z_j, Q_j)_j$. Then $\tilde{\Lambda}_\beta$ is continuous on $\mathcal{K}$, hence bounded by a constant C_β, and (13) follows from (15) with $C = C_2 \max_{|\beta|\leq m} C_\beta$. Note that $\tilde{w}$ is C^∞ on $\mathbb{R}^d \setminus \{0\}$, including

the sphere $|z| = 1$, because ψ is flat at 1; points with $|z_j| \geq 1$ have zero weight and can be added or removed freely.

(a) No point at the origin. This is the non-singular case of classical moving least squares [23, 15]. Let $\ell_{j,s}(\mathcal{C}) \in \mathbb{R}^{\dim \Pi_m}$ represent $\tilde{\lambda}_{j,s}$ in a fixed basis of Π_m; it is polynomial in z_j and linear in Q_j. The Gram matrix $G(\mathcal{C}) = \sum_j \tilde{w}(z_j) \sum_s \ell_{j,s} \ell_{j,s}^{\mathsf{T}}$ is C^∞ near $\mathcal{C}_0$, and it is positive definite at $\mathcal{C}_0$: a polynomial annihilated by all data with $\tilde{w}(z_j) > 0$ is annihilated by those with $|z_j| \leq c_0$, hence vanishes by (16). So G^{-1}, and with it the shape functions $\tilde{a}_{j,s} = e^{\mathsf{T}} G^{-1} \tilde{w}(z_j) \ell_{j,s}$ (with e the evaluation at 0), are C^∞ near $\mathcal{C}_0$.

(b) A point at the origin. By (H1), at most one point, say z_i, is near the origin. We repeat the proof of Theorem 4.1 with the configuration as the parameter in place of x. Write $\tilde{p}(z) = P(z - z_i)$, so that the data at z_i become $q_{i,s}(D)P(0)$ and do not depend on z_i, while all other data and the evaluation $\tilde{p}(0) = P(-z_i)$ depend polynomially on the configuration. Dividing the scaled functional by $\tilde{w}(z_i)$ gives $|\Lambda_i P - \tilde{y}_i|^2 + t\, B_{\mathcal{C}}(P)$ with $t = |z_i|^\alpha$, a polynomial in z_i, and $B_{\mathcal{C}}$ smooth in $\mathcal{C}$ near $\mathcal{C}_0$. The kernel N_i of Λ_i depends on Q_i, but since Λ_i has full rank it can be written, near Q_i^0, as the graph of a linear map from a fixed space that depends rationally on Q_i. The implicit function theorem then applies as in the proof of Theorem 4.1: the first diagonal block of the Jacobian is invertible because Q_i is independent, and the second, the Hessian of the frozen problem on N_i, is positive definite by (16), which involves the point $z_i = 0$. It yields a C^∞ function of $(\mathcal{C}, t)$. This function is the minimiser whenever $t > 0$: the map F is affine in the unknowns, its linear part is invertible near $(\mathcal{C}_0, 0)$, and for $t > 0$ the substitution $r = ts$ maps its zeros bijectively onto the critical points of the scaled functional; a positive semidefinite quadratic with exactly one critical point has a nonsingular, hence positive definite, Hessian, and that point is its unique minimiser; composed with $t = |z_i|^\alpha$ it gives the shape functions near $\mathcal{C}_0$, as C^∞ functions of $\mathcal{C}$. □

Remark 6.2.

(a) *Compact support.* The proof uses that only the finitely many nodes in $B(x, \delta)$ influence $u(x)$, so that the scaled configurations form a compact set. For weights with a Gaussian factor every node contributes, and an extension would need estimates of the tails; the experiments of Section 7 show the same rates for such weights.

(b) *Scaling of derivative data.* The factors $\delta^{2r_{j,s}}$ in (12) make the scaled problem independent of δ. Without them the relative weight of derivative data grows like δ^{-2r} as $\delta \to 0$, and the argument does not apply.

(c) *The norming condition.* If the data contain the function values, (H3) holds as soon as the value data alone satisfy it, since adding data can only increase the right-hand side. For values, local norming conditions of this type are the basis of the approximation theory of moving least squares on quasi-uniform nodes with δ a sufficiently large multiple of the fill distance [23, Chapters 3 and 4].

(d) *Gaps and completion.* The theorem applies to any data satisfying (H1)–(H3), with or without gaps, and so to completed data (Proposition 5.3) as well; the

added symbols can be taken homogeneous, since derivatives of homogeneous symbols are homogeneous. If the added values are only estimates, with errors η, then u changes by $\sum a_{j,s}\eta_{j,s}$, which by Step 3 is at most $\tilde{\Lambda}_0 \max_{j,s} \delta^{r_{j,s}}|\eta_{j,s}|$ in absolute value, where $\tilde{\Lambda}_0 \le C_0$ is the bound of Step 5.

7. Numerical experiments

The experiments test the theory of Sections 4 and 5 datum by datum. They use a numerically stable evaluator of SWIMLS, described in Appendix A, which stays accurate to rounding level as close to the nodes as we tested (10^{-14}). Whether a datum is reproduced is decided in high-precision arithmetic (Appendix B); the outcome is never borderline: reproduced data have errors below 10^{-18}, the others above 10^{-2}. Unless stated otherwise, nodes are random, the weights include a Gaussian factor (Appendix A), the data values are random numbers (exactness does not depend on them being samples of a function), and every node other than the one under study carries Hermite data. Scripts, logs and data are in the accompanying repository.

7.1. **Hermite and Hermite–Birkhoff data.** We first checked every datum at a node with classical Hermite data, of orders $k = 2$ to 7 in one, two and three dimensions and with α from 2 to 6, and at a node carrying the harmonic symbols of degree at most 3: 92 data in all. A datum is reproduced exactly when $\Delta^{\alpha/2} q = 0$ and not otherwise, as Corollary 4.5 predicts for the reproduced data and Proposition 4.7 and Conjecture 4.13 for the others. For instance, at $\alpha = 2$ in three dimensions, $\partial_x\partial_y\partial_z f$ is reproduced while $\partial_x^2 f$ has an error of 30. The error identities of Example 4.6 hold to all twelve digits computed: at a node with Hermite data of order 3 in two dimensions and $\alpha = 2$, $e(\partial_x^2) = e(\partial_y^2) = -45.904682196$ and $e(\partial_x^3) = 3e(\partial_x\partial_y^2) = 599.09999304$.

Next, one node carries a pattern with gaps, such as $\{f, \partial_{xx} f\}$, $\{f, \partial_x f, \partial_{xy} f\}$, $\{f, \Delta f\}$ or $\{\partial_x f\}$ alone: 23 patterns in one and two dimensions, most of them with gaps, plus the two settings of Section 7.3, 83 data in all. Again conditions (i) and (ii) predict every outcome. For example, $\partial_{xy} f$ is not reproduced in the pattern $\{f, \partial_x f, \partial_{xy} f\}$, because its derivative ξ_2 is missing from the node, and Δf is not reproduced in $\{f, \Delta f\}$.

7.2. **When condition (i) is not needed.** The configurations above are generic. Theorem 4.9 predicts that special ones behave differently, and they do. In each configuration of the following table condition (i) fails and yet the datum is reproduced; a small perturbation that destroys the special structure makes the error reappear.

configuration	datum	reason	error
1D: $f'(0)$, $f(1)$, $f'(5/2)$; $m=1$, $\alpha=2$	f' at 0, 5/2	Cor. 4.10(b)	$< 10^{-44}$
add $f(4)$			0.21–2.1
2D, $K=2$: $\nabla f(x_1)$; $f, \nabla f, \nabla^2 f$ at x_2; $m=2$, $\alpha=2$	∂_x, ∂_y	Cor. 4.10(a)	$< 10^{-39}$
add a third node			0.03–0.46
2D, $K=2$: $f, \partial_{xx} f$ at x_1; Hermite $k=2$ at x_2; $\alpha=4$	∂_{xx}	Cor. 4.10(a)	$< 10^{-28}$
add a third node			2.9–3.3
2D: $\partial_x f(0)$; others on the x_2-axis; $m=1$, $\alpha=2$	∂_x	Prop. 4.12	$< 10^{-37}$
move the others by 0.05			0.08–0.16

In the symmetric configuration (last two rows) the weights of the other nodes are even in x_1, so $\partial_{x_1} Z(x_i) = 0$ and the first-order error (10) vanishes for $q = \xi_1$. The restriction $\deg q \le \alpha$ in Theorem 4.9 also matters: with two nodes carrying Hermite data of order 3 and $\alpha = 2$, every datum with $\Delta q \neq 0$ fails, with errors between 4.9 and 83.

7.3. **Derivatives at points without values.** In the setting of [8, §3], eight random points carry values, eight other points carry gradients, the weights are $|x - x_i|^{-\alpha}$ and the local space is Π_1 or Π_2. For $m \in \{1, 2\}$ and $\alpha \in \{2, 4\}$, none of the 24 gradient components checked is reproduced; the errors lie between 0.06 and 4.5. After completion (Proposition 5.3) they are, and completion also repairs all 13 non-reproduced data of the patterns of Section 7.1. Exactness is not free in accuracy, though: with values at 1024 Halton points and gradients at 1024 others for Franke's function, the maximum error is $1.2 \cdot 10^{-3}$ without and $3.4 \cdot 10^{-3}$ with completion, because the added values are only estimates (here from a simple least-squares fit).

7.4. **Accuracy.** We have no convergence theory for SWIMLS with derivative data, and report a parameter study instead: Halton nodes in the unit square ($N \le 4096$), a smooth test function, Gaussian factors of width proportional to $h = N^{-1/2}$ in the weights, and 44 combinations of the data order $k \le 2$, the degree $m \le 5$, α and the width. Over the last three refinements the root-mean-square error decays like h^r with $|r - (m+1)| \le 0.53$ in every case, the rate known for MLS approximation with local degree m [10, 23]. Figure 2 shows what derivative data buy. On a fixed node set they reduce the error, e.g. for $N = 4096$ and $m = 5$ from $5.0 \cdot 10^{-9}$ to $1.1 \cdot 10^{-9}$ with gradients and Hessians. Per number of data, however, values alone were more accurate in this test: derivative data pay off when nodes are expensive and derivatives are cheap, as with adjoint gradients.

8. Conclusions

Singular-weight moving least squares extends from function values to arbitrary linear differential data with no change to its definition: one weighted least-squares problem per evaluation point, with no mesh, no global system and no poisedness condition beyond unisolvency of all the data on the local polynomial space. The interpolant is C^∞, and Theorem 4.3 accounts exactly for the error in every datum. Two mechanisms

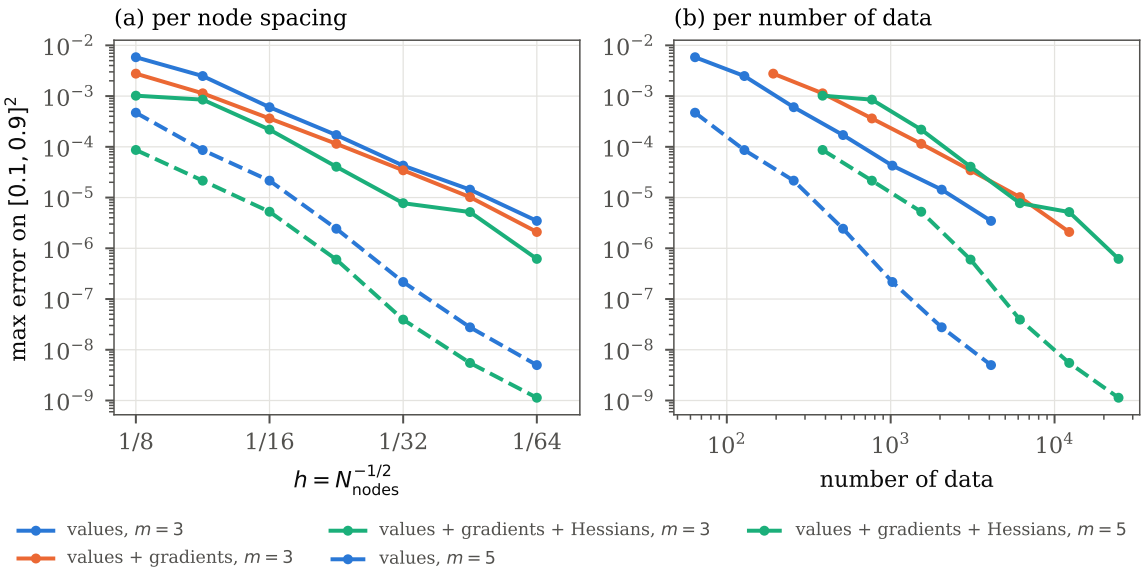


FIGURE 2. Maximum error on $[0.1, 0.9]^2$ for Halton nodes, $\delta = 2h$; $\alpha = 2$ except for Hessian data ($\alpha = 4$). (a) Against the node spacing. (b) Against the total number of data.

can spoil a datum: the singularity of the weight, which acts through $\Delta^{\alpha/2}$, and the motion of the frozen minimiser, which acts only on data whose derivatives are missing at their node. The first is removed by choosing α larger than the order of the data, the second by completing the data at the node. Together they give a simple recipe for Hermite–Birkhoff interpolation of any order in any dimension.

Several questions remain open. Conjecture 4.13 asks whether $\Delta^{\alpha/2} q \neq 0$ always causes an error in generic configurations; we proved it at leading order. Beyond the first-order formula (10), we do not know exactly which configurations of the other nodes make condition (i) unnecessary. The convergence orders observed in Section 7.4 call for an error analysis in the spirit of [10, 23] that accounts for derivative data and singular weights. Finally, the values added by completion affect the accuracy, not the exactness, and better ways of estimating them would reduce the accuracy penalty observed in Section 7.

APPENDIX A. THE EVALUATOR

The usual implementation of interpolating MLS replaces $|x - x_i|^{-\alpha}$ by $(|x - x_i|^{\alpha} + \varepsilon)^{-1}$ and solves the normal equations. This gives up exactness at the nodes and, near a node, squares the condition number of the weighted system, which grows like $|x - x_i|^{-\alpha/2}$. Our evaluator avoids both problems; it is in the accompanying code.

A.1. **Basis and weights.** At the evaluation point x we use the monomials $((y - x)/h)^{\beta}$, $|\beta| \leq m$, with a length scale h, so that $u(x)$ is the first coefficient. The weights are

either pure powers $|x-x_i|^{-\alpha}$ or $w_i(x)=|x-x_i|^{-\alpha}e^{-|x-x_i|^2/\delta^2}$, which also satisfy (A1); they are computed as logarithms and normalised by their maximum, and nodes whose normalised weight is below a cutoff are dropped. The cutoff is measured relative to the strongest node *other than* the nearest one; measured relative to the nearest node, it would drop all other nodes close to a node and destroy the information that fixes ν. Rows of derivative data of order r are multiplied by ℓ^r for a length ℓ; by the remark after Definition 3.6 this does not affect the results of Section 4.

A.2. **Least squares.** The weighted system $W^{1/2}A\,a \approx W^{1/2}b$ is solved by an SVD-based least-squares solver rather than through the normal equations. Close to a node x_i the rows of x_i dominate the others by a factor s^{-1} with $s\sim|x-x_i|^{\alpha/2}$. The light rows carry the information that fixes $\nu(x)$, and a standard solver resolves them only to relative accuracy of order $\varepsilon_{\mathrm{mach}}/s$ (and not at all through the normal equations once $s^2<\varepsilon_{\mathrm{mach}}$). When $s<10^{-3}$ we therefore solve an exact reformulation. Let R and L be the heavy and light rows, Q_1 and Q_2 orthonormal bases of the row space and the null space of R, and c_0 the least-squares solution of $RQ_1c=b_R$. Writing $a=Q_1(c_0+s\,d)+Q_2z$ and dividing the objective by s^2 gives the problem

$$\min_{d,z}\ \|RQ_1d\|^2+\left\|LQ_1d+s^{-1}LQ_2z-s^{-1}(b_L-LQ_1c_0)\right\|^2,$$

whose blocks are well scaled; it has the same minimiser and is the discrete counterpart of the decomposition (5).

A.3. **At the nodes.** At $x=x_i$, and within a tolerance of 10^{-13} times the size of the node set, we use the limit of Theorem 4.1: the data at x_i are imposed as equality constraints, and the remaining freedom in N_i is fixed by minimising the frozen functional (6); then $u(x_i)=(p_i^*+Z(x_i))(x_i)$.

A.4. **Accuracy.** Figure 3 compares the evaluator with a reference computed with 120 significant digits, as the evaluation point approaches a node. Its error stays at the level of rounding (below $5\cdot10^{-15}$) down to distance 10^{-14}; the three smallest distances are within the node tolerance and use the formula at the node. With $\varepsilon=10^{-10}$ the regularised normal equations have errors up to $1.6\cdot10^{-10}$, the size of the regularisation; with $\varepsilon=10^{-16}$ the errors reach $4\cdot10^{-12}$ and the matrix is numerically singular at some distances.

Appendix B. Verifying exactness

The experiments of Section 7 decide whether a datum q is reproduced by computing $\big[q,u\big]_{x_i}$ in high-precision arithmetic (`mpmath`). The interpolant is evaluated from the normal equations at a working precision of $40+(2\alpha+|\beta|+5)\,|\log_{10}r|$ digits, and $D^\beta u(x_i)$ is taken with a tensor-product Chebyshev stencil of radius r in the variables with $\beta_j>0$, which is exact for polynomials of degree at least β_j+5 in each of them and never samples the node itself. A datum is called reproduced if $\big|\big[q,u\big]_{x_i}-y_q\big|\le10^{-12}\max(1,|y_q|)$ for

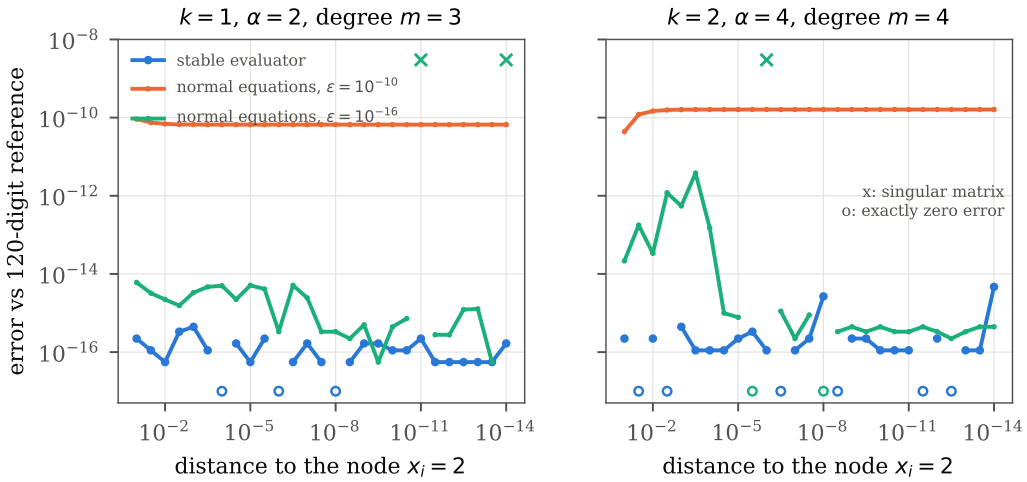


FIGURE 3. Error of the evaluation near the node $x_i = 2$ (nodes $1, 2, 3, 4$ with Hermite data of order k and random values), against a 120-digit reference. Crosses mark singular normal equations, open circles errors that are exactly zero.

both $r = 10^{-6}$ and $r = 10^{-5}$, and not reproduced if the inequality fails for both. Mixed outcomes never occurred, with a wide margin: over the 221 data of Sections 7.1–7.2, the reproduced ones have errors between $6 \cdot 10^{-55}$ and $9 \cdot 10^{-25}$ at $r = 10^{-6}$ (at most $9 \cdot 10^{-19}$ at $r = 10^{-5}$), and their errors scale like r^6 or r^8, as the stencil error does; the others have errors of at least 0.01, identical at both radii.

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